The Evolution of Transit Access to Essential Services

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Draft: November 12, 2026

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Abstract

This paper measures how neighborhood transit access to five essential services changed and whether the transit network or the destinations moved. We contribute an original dataset: quarterly transit access to five essential services for every block group in 52 U.S. metropolitan areas over 2014–2025, scored on travel-time matrices routed from archived transit schedules with one fixed method, the same matrices that measure job access. Each service's within-neighborhood change splits exactly into transit-network and destination components in the 19 metros with the most complete archives. The mechanisms differ by service, and the network reduced access to every service after 2019. Bank access fell 18.1 percent, driven by branch closures; library access fell 2.5 percent, driven by the network; hospital access fell 3.4 percent, split between the two sides; grocery access roughly held (up 1.7 percent) as store growth offset network decline; community colleges were flat once reclassified and consolidated colleges are not mistaken for closures. The metro-clustered intervals for the bank and hospital declines exclude zero; the smaller changes' intervals do not. Under the archive-entry correction, declines were largest in majority-Black neighborhoods, which started with more access than majority-white ones. Deep losses compound, mostly where the transit network declined. Neighborhoods with declines in at least four of six services, jobs included, are 5.5 times as common as independent declines would imply, and those with five or more are 31.5 times as common. These neighborhoods are poorer and more carless, with larger minority shares.

Keywords: accessibility; public transit; essential services; transport equity; healthcare access; social exclusion

*Xavier University. Contact: arnolds7@xavier.edu. Transit schedules include data from Transitland (https://www.transit.land/terms). Street network data © OpenStreetMap contributors, available under the Open Database License (https://www.openstreetmap.org/copyright). Routing was performed on resources provided by the Ohio Supercomputer Center (Ohio Supercomputer Center, 1987, Columbus, OH, https://ror.org/01apna436), whose support we gratefully acknowledge.

1 Introduction

A household without a car reaches the bank, the grocery store, the library, the hospital, and the community college by transit, the same way it reaches work. Nationally, about 8% of households lack a vehicle, and 19% of lower-income and 21% of Black adults seldom or never drive, against 3% of upper-income and 7% of White adults (Pew Research Center, 2024; U.S. Census Bureau, 2023b).

In urban food deserts, the exit of transit service is associated with fewer grocery trips by lower-income households and less healthy food purchases (Arnold, 2025). Bank branches still matter to households and businesses even as consolidation removed nearly a fifth of them (Adams and Sherlund, 2025). In the 2004 Public Libraries Survey, library systems in lower-income and rural neighborhoods were relatively less funded and offered fewer information resources (Sin, 2011). College-going is largely local, with most students enrolling near home, so a community with no college nearby is an education desert (Hillman, 2016). Spatial access to care is the organizing concern of a large healthcare-access literature (Neutens, 2015; Buchalter et al., 2025). A neighborhood that loses access to several of these services at once shows the lack of access to essential goods and services that the transport-poverty literature links to social exclusion (Lucas, 2012).

Over 2014–2025 both the destinations and the transit networks changed. Branches consolidated; supermarkets and hospitals opened and closed; and agencies cut service as ridership fell during the pandemic (Kar et al., 2022). This paper asks four questions about that period. How did transit access to each essential service change? How much of each change came from the transit network and how much from the destinations? Which neighborhoods, by race, income, and car ownership, had the largest declines? Did the same neighborhoods lose access to several services at once? The second question matters for policy. A transit agency cannot reopen a closed branch, and a banking regulator cannot redesign a bus network. Attributing a decline to one side requires following both sides over time.

Accessibility research has mostly measured destination types one at a time (Section 2; Kelobonye et al., 2019), so relatively little is known about whether different essential services have changed in the same ways over time. We contribute an original dataset for this comparison: quarterly transit access to five essential services for every block group in 52 U.S. metropolitan areas over 2014–2025, measured identically throughout, with job access alongside for comparison. The routed travel times and the network-versus-destination decomposition come from Arnold (2026); this paper adds the five destination layers and applies both to a new question.

There are four main results. First, trends differ by service. Bank access fell 18.1% over

2014–2025 while grocery access rose 1.7%, with libraries, hospitals, and community colleges in between. Second, the mechanisms differ even more. Bank access fell mainly through branch closures, library access through the network, and hospital access through both. Destination change differs by service, while the network side declined for every service after 2019. Third, after correcting for changes in archive coverage, majority-Black neighborhoods, which started with more access than majority-white ones, had the largest declines in every essential service, and the poorest and most carless neighborhoods, which also started with more access than the richest and most car-owning, had larger declines in nearly all of them. Fourth, losses compound across services. Declines of about 5 percent or more in four or more services occur in 5.5 times as many neighborhoods as independent declines would imply, and in five or more services, 31.5 times as many. Among neighborhoods with similar network change, the two ratios fall to 1.08 and 1.33, so most of the excess comes from network declines that reduced access to several services in the same neighborhoods. These neighborhoods are poorer, more carless, and more central, with larger minority shares. The claims are descriptive throughout and concern access itself, not its effects on outcomes.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the travel-time matrices and the destination layers, and Section 4 defines the access measure and its decomposition. Section 5 reports each service's trajectory and decomposition. Section 6 compares the six services, identifies which neighborhoods had the largest declines, and tests whether losses compound across services. Section 7 reports New York separately. Section 8 concludes, and Appendix A holds the robustness and validation exhibits.

2 Related literature

This paper draws on three literatures. The first is accessibility measurement. Accessibility measured as the potential for reaching opportunities originates with Hansen (1959), and its cumulative-opportunity form with Wachs and Kumagai (1973); Handy and Niemeier (1997) and Geurs and van Wee (2004) survey the taxonomy and its trade-offs. The health-geography literature measures access to care with floating-catchment and gravity constructs (Luo and Wang, 2003), and Cui and Levinson (2020) formalize the primal/dual distinction. Karner (2018) builds route-level accessibility measures from published transit schedules to evaluate transit service equity. Longitudinal measurement exists for jobs. The *Access Across America* series scores transit access to employment annually across roughly fifty metros (Owen et al., 2025), and Foth et al. (2013) track access against social need across two census years in Toronto. Boisjoly and El-Geneidy (2016) find that a constant 8 a.m. measure of job

accessibility in Toronto is representative of relative accessibility across the day. Farber and Fu (2017) compare job access in Portland and the Wasatch Front before and after service changes, holding jobs at their 2011 locations so that the measured change comes from the transit network alone. These studies established repeated measurement, but none follows one fixed method over a long panel or splits each change into network and destination parts. Multi-service measurement also exists. England's Journey Time Statistics report travel times to employment centers and seven other types of key local service, and their 2019 release states that changes to data and methodology mean that "robust conclusions cannot be made about changes over time" (Department for Transport, 2021). The TransitCenter Equity Dashboard (Klumpenhouwer et al., 2021) tracks routed transit access to healthcare and groceries in seven regions from 2020 on, and Li et al. (2025) compute one-date two-step floating catchment (2SFCA) indices for several point-of-interest classes in five cities. Kar et al. (2022) compare block-group transit access to groceries and health care across three phases of 2020 in 22 U.S. cities, with destinations fixed at 2019 locations. In their data, block groups where several social disadvantages coincided were more likely to lose access, which is compounding across disadvantages rather than across services (compare Section 6.3).

The second literature concerns the components of access. The conceptual split of access change into transport and land-use parts is in Geurs and van Wee (2004); longitudinal decompositions exist for European road accessibility (Stępniak and Rosik, 2018; Rosik et al., 2025), and Zhang et al. (2026) follow New York City block groups across four snapshots (2014–2022), separating transit-network change from residential sorting in a widening racial access gap. In the U.S., Janatabadi and Ermagun established a framework for untangling the place and mobility components of access at a point in time. In their framework, holding destinations counterfactually uniform isolates the contribution of transit service to spatial inequality. They apply it to employment access in the Washington region and the City of Chicago (Janatabadi and Ermagun, 2024a,b). In our terms, their "mobility" or "transit service" side is the network component and their "place" or "land use" side the destination component. Ermagun and Witlox (2024) compared fifty metropolitan statistical areas, and Ermagun et al. (2025) measured fine-grained, time-denominated (nearest-destination) access to six destination types across Chicago, showing that which service binds varies sharply across neighborhoods.

The third is the set of service-domain literatures, each with its own convention. For banking, Adams and Sherlund (2025) measure block-level distance to the nearest branch through the consolidation decade and find it moved little. For healthcare, Neutens (2015) reviews the accessibility tradition for transport geographers. Buchalter et al. (2025) build a multi-modal composite across 56 provider classes from quarterly 2016–2020 provider files,

with transit times routed on feeds from an open feed database. Food access has the U.S. Department of Agriculture (USDA) food-desert measures (Ver Ploeg et al., 2009) and a transit-specific literature measuring supermarket access within a city as it varies over the day (Farber et al., 2014) and with commuters' space-time constraints (Widener et al., 2015). Education has "education deserts", counts of colleges within commuting zones (Hillman, 2016), and libraries are benchmarked by average distance to the nearest outlet (Donnelly, 2015). On the transit side, Jomehpour Chahar Aman and Smith-Colin (2020) identify transit deserts by overlaying transit supply on transit-dependent demand. We depart from these distance- and area-based conventions (Section 4) and report nearest-facility levels as context.

The motivating frame is the transport-poverty and social-exclusion literature (Lucas, 2012; Pereira et al., 2017) rather than spatial mismatch. Martens (2012) supplies its normative reading, in which accessibility is the transport good whose distribution is at stake. The equity questions of Section 6.2 follow the U.S. work on nonwork accessibility by Grengs (2015) and the activity-participation reading of transport poverty in Allen and Farber (2020).

This paper adds to these literatures. To longitudinal measurement it adds an original dataset of five essential services measured with the differencing design of Arnold (2026): one fixed method applied to a 2013–2025 archive and corrected for archive-coverage changes. To the components literature it adds network-versus-destination attributions of change for the same services, comparable across metros and services. To the service-domain literatures it adds six services measured identically, which makes it possible to measure the overlap of losses across services.

3 Data

This section describes the dataset that the paper contributes. Its travel times are the routed panel described in Section 3.1, and its five destination layers, described in Section 3.2, are original to this paper. Scoring each layer on the travel times gives quarterly transit access to five essential services for every block group in the covered metros over 2014–2025.

3.1 Travel-time matrices (shared infrastructure)

The routed panel is that of Arnold (2026): archived schedules in the General Transit Feed Specification (GTFS) for every covered metro, one sample date per quarter, routed door-to-door between 2020-vintage census block-group centroids on OpenStreetMap networks with the R5 routing engine (Conway et al., 2018). Each pair's value is the median travel time, including walking, waiting, in-vehicle time, and transfers, over departures in the 7:00–9:00

morning peak. Trips longer than 90 minutes are treated as out of reach, and all other routing parameters are engine defaults, held fixed across metros and quarters. Centroid-to-centroid routing introduces travel-time error in large peripheral block groups. The centroids are fixed, so this error is constant over time, although under a fixed travel-time threshold a constant error can still change which network changes cross that threshold. The archive spans 2013–2025; the analysis window, and every quoted change, runs 2014–2025 (48 quarters). When an agency's feeds enter or leave the archive, measured access can change in that quarter with no change in service, and Section 4 removes these changes. In four metros a rail service leaves the archived feeds while the set of feeds stays the same, so that removal does not apply. Seattle's feeds omit Link light rail from 2021Q3. Sacramento's omit its light rail from 2020Q3 to 2023Q4, and the rail returns in 2024Q1 at a change in the set of feeds, so the removal also takes out the return and the loss persists through 2025. Dallas's and Austin's feeds omit their rail lines for two quarters of 2014, and Austin's again in 2023Q3. Los Angeles's feeds omit Metro Rail from 2016 on, at a change in the set of feeds, so the removal applies and only the level of its access is understated. Feed-quality flags (outage and thin-feed quarters) carry over; flagged metro-quarters are excluded from the estimates, and each trajectory is also reported with them included. The balanced panel covers 52 metros outside New York (Section 3.3; Appendix Table A.1; supplementary Table S1); New York (NYC) is reported separately throughout because its gradients differ qualitatively (Section 7).

3.2 Destination layers

Each service comes from a public administrative census of facilities (Federal Deposit Insurance Corporation, 2025; Institute of Museum and Library Services, 2026; U.S. Department of Agriculture, Food and Nutrition Administration, 2026; National Center for Education Statistics, 2025; Centers for Medicare and Medicaid Services, 2026), geocoded and aggregated to 2020 block groups, held constant within calendar years so that sub-annual access movement is network-driven by construction. The library and community-college files end in 2024 and are held for 2025. Table 1 summarizes.

For banks, the lead counts full-service offices (96–97% of all offices), as Adams and Sherlund (2025) do. For groceries, the lead is full-line food retail (supermarkets, superstores, large grocery stores), the USDA food-desert family (Ver Ploeg et al., 2009); all SNAP-authorized retailers, including convenience and dollar stores, form a broader variant (Section 5.3). For community colleges, two artifacts both lower the raw count. IPEDS sector codes reclassify community colleges that begin awarding bachelor's degrees (140 of the 1,047 public two-year institutions in the 2013 file had been reclassified by 2024), and consolidations merge

Table 1: Destination layers.

Service	Source	Definition (lead)	2014	Latest
Bank branches	FDIC Summary of Deposits	full-service branches	89,673	73,616
Libraries	IMLS Public Libraries Survey	stationary outlets (CE+BR)	16,906	16,864
Groceries	USDA SNAP retailer history	supermarkets, superstores, large grocery stores	42,110	44,027
Community colleges	NCES IPEDS	persistent-campus public 2-year	1,045	1,065
Hospitals	CMS Provider of Services (Dec. vintages)	durable general acute care (geocoded; 97–98% coverage)	4,694	4,383

Note: FDIC = Federal Deposit Insurance Corporation; IMLS = Institute of Museum and Library Services; CE+BR = central and branch libraries; USDA SNAP = U.S. Department of Agriculture, Supplemental Nutrition Assistance Program; NCES IPEDS = National Center for Education Statistics, Integrated Postsecondary Education Data System; CMS = Centers for Medicare and Medicaid Services. Lead = the definition behind every headline result; variants are robustness checks. National counts of the geocoded lead layers, totals over the fifty states, DC, and the territories each source covers; latest = 2025, or 2024 for libraries and community colleges. In 2023 the 92 metros of the analysis panel (Section 3.3) hold 35–55% of each layer's national facilities (banks 51%, libraries 38%, groceries 55%, community colleges 35%, hospitals 35%), and the 52 panel metros hold 23–36%.

multi-campus systems into single records while the campuses keep operating (48 of the 64 institutions that left the file by 2024). The lead, the persistent-campus definition, therefore assigns each institution a persistent identity and carries a consolidated system's campuses forward on IPEDS's own merger crosswalk (53 campuses across 22 systems; the persistent-identity variant omits the carry-forward); Section 5.4 quantifies the decline the raw categories would show. For hospitals, CMS certified hundreds of temporary facilities during the pandemic (264 new general-acute certifications in 2020 alone, 225 of them gone by December 2023). The lead counts durable general acute hospitals (short-term acute and critical access), certifications active in at least three consecutive December vintages, with facilities present in 2013 and 2024–25 entrants retained; 184 of 8,440 certifications are excluded as transient. The rule removes only the shortest-lived of the 2020 cohort (48 of 264; Section 5.5). The healthcare object is hospitals because we found no public site-level history of community health centers before 2021. We audit every service's classification codes and reporting units for these artifacts; groceries' re-typing flows are small and symmetric, and banks' and libraries' codes are stable within facility.

Lead hospitals resolve to coordinates at 97–98% per year (91% for all hospitals; Appendix Table A.2). The ungeocoded set is persistent rather than churning (91.2% of all hospital certifications geocode in every year they appear, and 1.1% ever switch status), so within-neighborhood trends compare the same facilities over time, and levels are understated by roughly the coverage gap.

The grocery layer reconciles against an independent count (Census County Business

Patterns; U.S. Census Bureau, 2025). The hospital layer's state counts correlate at 0.997 with the Homeland Infrastructure Foundation-Level Data (HIFLD) facility layer (Homeland Infrastructure Foundation-Level Data, 2020), which also supplied some of its coordinates (Appendix Table A.3). The national totals in Table 1 moved in the same direction as the covered metros' facility counts for banks, groceries, community colleges, and hospitals. Library outlets were flat nationally but rose in the covered metros (Section 5.2).

3.3 How representative are the 52 metros?

The balanced panel, called the panel from here on, is the set of metros whose archived schedules support routing in at least 36 of the 48 quarters from 2014 through 2025. The 52 balanced metros outside New York hold roughly 45% of U.S. metropolitan population and include 30 of the 50 largest metros. Of the 96 routed metros, 92 enter the unbalanced analysis panel and 90 (189 million residents) carry census attributes (Appendix Table A.1). Several legacy-transit metros fall below the balance bar because their schedules were missing from the archive or unavailable to this build: San Francisco, Minneapolis, Baltimore, Pittsburgh, St. Louis, and Detroit. Washington, DC is present, but its principal operator, the Washington Metropolitan Area Transit Authority (WMATA; Metrorail and Metrobus), is missing throughout, so its routed access is understated. Appendix Table A.3 reports the bank and hospital trajectories without it. Archive coverage favors larger, better-archived agencies. We therefore read the results as describing transit access in these metros, not in the average U.S. metropolitan area.

4 Measuring access and decomposing its change

4.1 Access measure

Access A for block group i, service s, quarter t is the count of s-facilities reachable from i within τ minutes by transit and walking, with $\tau = 45$ as the primary budget; in the full 15/30/45/60 curve (Appendix Table A.4) every declining trajectory grows in magnitude with the budget. One construct serves as both trajectory and decomposition object for every service, and the domain conventions are carried as robustness variants. Cumulative and gravity measures of job access correlate above 0.9 when the cumulative threshold is set at the mean commute time (Santana Palacios and El-Geneidy, 2022). Among the construct variants of Appendix Table A.5, the only one that reverses a trajectory's sign is community-college coverage, the share of residents reaching at least one campus within 45 minutes (+0.5 percentage points against a -0.4% trajectory).

Departure time is the one convention this measure cannot vary without new routing. Trips to banks, libraries, groceries, and hospitals are not limited to the morning peak, and transit access to supermarkets varies within the day (Farber et al., 2014). Across 22 U.S. cities in 2020, changes in transit access to groceries and health care differed little between peak and off-peak hours (Kar et al., 2022). After 2020, all-day service recovered faster than the network component of peak-period job access in the 19 metros used for this paper's decompositions (Arnold, 2026), so the network components reported here, which describe morning-peak access only, may overstate the post-2019 decline for trips made outside the peak.

All trend estimates are within-neighborhood. We regress log access (plus one) on quarter effects with a separate fixed effect for each block group in each span of quarters over which its metro's set of archived feeds is unchanged, block-group population weights (the fixed 2018–22 American Community Survey (ACS) vintage; U.S. Census Bureau, 2023a), and standard errors clustered by block group. Composition changes and changes in archive coverage therefore do not appear as access changes. Quoted endpoints compare the 2025 annual mean of quarter effects to the 2014 annual base, and seasonality is small (Appendix Table A.3). The block-group-clustered intervals quoted throughout describe the panel, while the metro-clustered intervals reported with each trajectory are the test for any claim that a change is common across metros like these.

4.2 Decomposition

For each metro we fix the base network n_b (the first unflagged 2014 quarter) and the base destination vector d_b (2014), and compute access $A(n, d)$ under all four combinations of network (n_b or a later n_t) and destinations (d_b or d_t). The observed within-neighborhood log change splits exactly and symmetrically (a two-factor Shapley decomposition; Shorrocks, 2013):

$$\underbrace{\ln A(n_t, d_t) - \ln A(n_b, d_b)}_{\text{observed}} = \underbrace{\tfrac{1}{2}\left[\Delta_n \mid d_b\right] + \tfrac{1}{2}\left[\Delta_n \mid d_t\right]}_{\text{network}} + \underbrace{\tfrac{1}{2}\left[\Delta_d \mid n_b\right] + \tfrac{1}{2}\left[\Delta_d \mid n_t\right]}_{\text{destination}}, \qquad (1)$$

where $\Delta_n \mid d$ is the change in $\ln A$ from n_b to n_t with destinations held at d, and $\Delta_d \mid n$ the change from d_b to d_t with the network held at n. The destination part is further split into a geometry-free scale term and a residual relocation term. Scale for metro m in year y is $\ln(T_{my}/T_{m,2014})$, the log change in the metro's total facility count on the fixed block-group universe. Relocation is the destination component minus scale: the part of the destination effect attributable to where facilities sit relative to the network rather than to how many

exist. The term is broader than facilities physically moving. It also includes the residents who reach no facility of the service, whose destination component is zero, so that their relocation term equals minus the scale term. Block-group changes are computed first and population-weighted averages second.

All access logs, including those in equation (1), are $\ln(A+1)$, so zero counts enter at zero and proportional changes are compressed for sparse services. Decomposition components are reported in log points, 100 times the change in $\ln(A+1)$, because they add up exactly; trajectories are reported as percent changes, and a change of x log points is $100[\exp(x/100)-1]$ percent.

The job-access layer is that of Arnold (2026), workplace employment from the Census Bureau's Longitudinal Employer–Household Dynamics (LEHD) Origin–Destination Employment Statistics (LODES; all jobs; U.S. Census Bureau, 2026), scored on the same matrices and decomposed by the same split. The latest LODES year is 2023, so the 2024 and 2025 quarters use 2023 job counts. Alaska's files end in 2016 and Michigan's in 2021, so Anchorage and Grand Rapids use their last available year after those dates.

The archive-entry correction removes from each block group's observed and network series the cumulative step in access at each change in its metro's set of archived feeds since the 2014 base, so the decomposition stays exact. Each step is the block group's change in log access between the last observed quarter before and the first observed quarter after that change. Appendix Figure A.1 shows the correction in one metro, and Appendix Table A.3 compares it with agency-reported revenue miles at the twelve archive-entry events, the changes in the set of archived feeds that move access by 3 log points or more. At those events the median absolute change in agency-reported revenue miles was 2.3 log points, against 3.0 at other metro-quarters, and the size of the access step is unrelated to the change in revenue miles (rank correlation -0.08), consistent with steps that reflect archive coverage rather than service. Every result outside Section 7 uses the corrected series. Because archive entry adds measured access on net, the uncorrected series is a diagnostic upper bound in aggregate.

The decompositions use every-year block groups, those observed and unflagged in every year from 2014 to 2025. The clean-archive core is the every-year block groups of the 19 metros that are never flagged and are observed in every year (28,147 block groups), and the all-metro companion is the every-year block groups of all 35 panel metros that have any (55,359). Appendix Table A.6 lists the quality flags that define the excluded metros. Excluding the flagged metros changes levels but, except for hospitals (Section 5.5), not mechanisms. Appendix Figure A.2 compares the component splits in the two samples. Excluding the five metros with rail gaps in their archived feeds (Section 3) moves each essential service's panel trajectory by at most 0.6 percentage points and leaves every mechanism in the clean-archive

core unchanged (Appendix Table A.3).

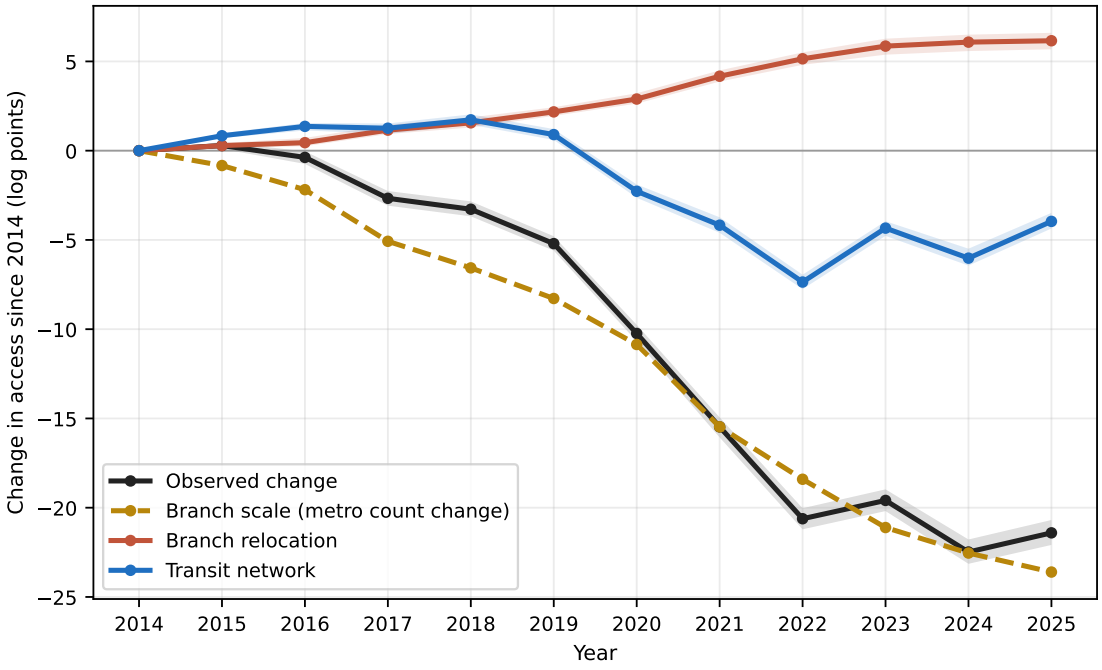
5 The evolution of access, service by service

Trajectories run on the panel and decompositions on the clean-archive core, with the all-metro companion beside every core decomposition; every exhibit names its sample (Appendix Table A.7).

5.1 Banks

Banks are the destination-driven service. In Figure 1, the observed −21.4 log-point change at the 2025 endpoint splits into −4.0 network, −23.6 branch scale, and +6.2 relocation (all-metro companion: $-17.9 = -3.2 - 19.9 + 5.2$). Branch closures, not transit change, removed about a sixth of the branches the typical neighborhood could reach within 45 minutes (a destination component of −17.4 log points).

Figure 1: Decomposition of 45-minute transit access to bank branches, 2014–2025.



Note: Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

The positive relocation term does not mean that closures avoided accessible places. It comes from the 23% of residents who reached no branch in 2014, whose access could not fall.

Among residents who reached at least one branch, the relocation term is slightly negative (−0.5 log points), consistent with closure rates that were mildly higher in the most transit-accessible quartile of 2014 locations than in the least (4.5 percentage points, standard error 0.8; Table 2).

Table 2: Branch closures by 2014 access quartile.

2014 branch location	Branches	Closed by 2025
Q1 (least transit access)	7,191	31.0%
Q2	7,191	31.4%
Q3	7,190	31.3%
Q4 (most transit access)	7,191	35.5%

Note: Full-service branches active in 2014 within the panel's block groups, quartiles of the location's 2014 45-minute transit count, and the share closed by 2025.

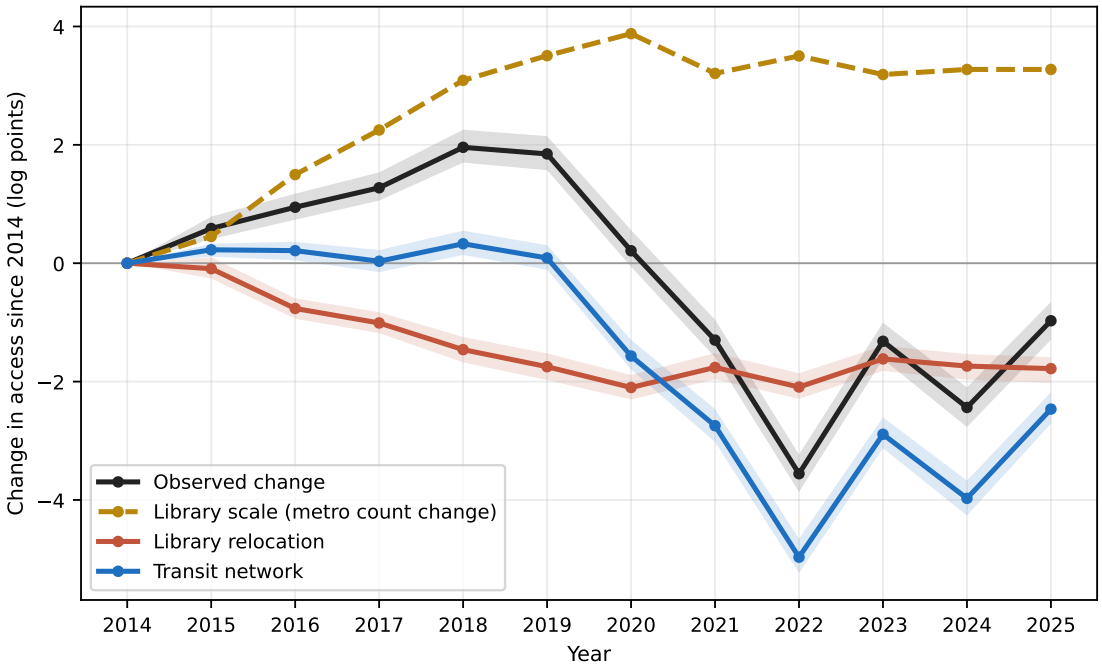
In trajectory terms, transit access to full-service branches fell 18.1% [95% CI: −18.6, −17.6] (flags included: −16.1%; all offices: −18.4%; NYC included: −18.7%). The metro-clustered interval also excludes zero ([−22.5, −13.3]). Moving from the fixed-effects trajectory on the panel (−19.9 log points) to annual means on the panel's every-year block groups gives −17.9, and restricting those block groups to the clean-archive core gives the decomposition's −21.4. Distance to the nearest branch barely changed over the same period (Adams and Sherlund, 2025), and our nearest-facility variant rises by 2 minutes (Appendix Table A.5), while the number of reachable branches fell.

5.2 Libraries

Libraries are the network-driven service. In Figure 2 the observed line moves with the network line after 2019. The observed −1.0 splits into −2.5 network, +3.3 scale, and −1.8 relocation (all-metro companion: $-1.6 = -2.4 + 1.9 - 1.1$). The network component exceeds the whole decline, while outlet growth (+3.3) more than offsets the adverse relocation (−1.8), about a fifth of which comes from residents who reach no outlet.

The panel trajectory for stationary outlets fell 2.5% [−2.8, −2.2] (flags included: −1.0%; with bookmobiles: identical; NYC included: −3.0%). The trajectory spans zero under metro clustering ([−5.1, +0.1]).

Figure 2: Decomposition of 45-minute transit access to library outlets, 2014–2025.



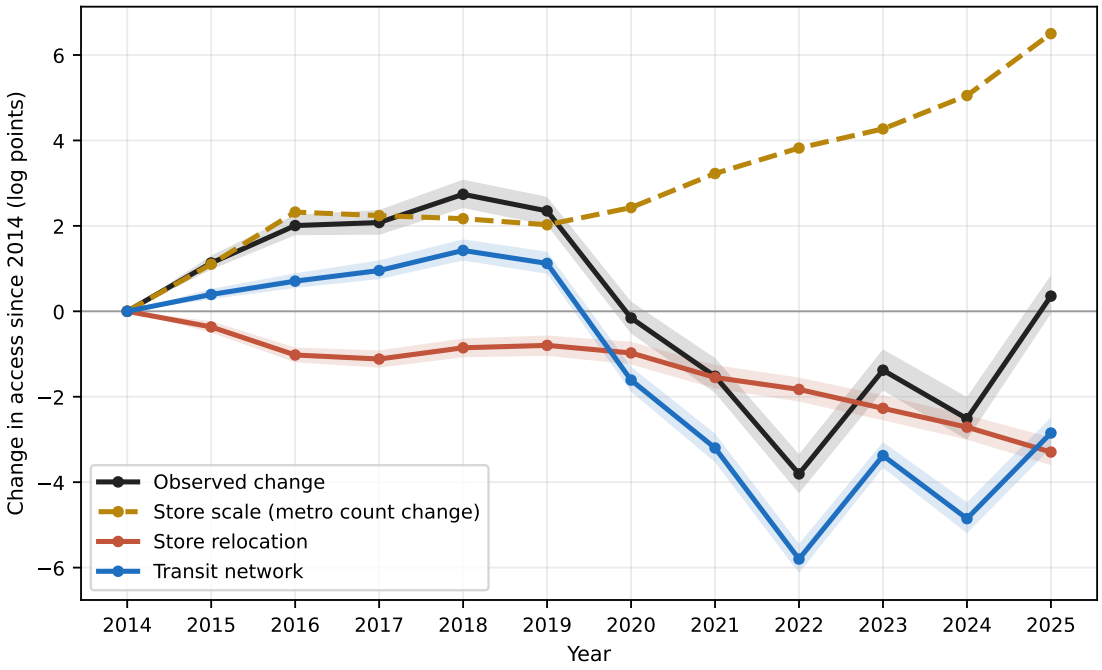
Note: Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

5.3 Groceries

Groceries show a third mechanism. In Figure 3 the store-scale component grows while the network declines. The negative relocation term is not read as stores moving, because it mostly reflects the +1 in $\ln(A+1)$ where few stores are reachable. The net change is small in both samples. The all-metro observed change is +1.9 ($= -2.7 + 8.4 - 3.8$), and in the clean-archive core it is +0.4 ($= -2.8 + 6.5 - 3.3$; 95% block-group bootstrap interval −0.1 to +0.8).

On the panel, transit access to full-line grocery stores rose 1.7% [+1.3, +2.2] (flags included: +3.8%; NYC included: +1.4%). Groceries are the only service whose panel-wide trajectory rises, though the metro-clustered interval spans zero ([−2.7, +6.4]). The all-SNAP variant is flat (−0.2%), so the gain is specific to full-line stores. The trajectory and the core decomposition endpoint differ in the same two steps as for banks (Section 5.1), from +1.7 log points to +1.9 and then +0.4.

Figure 3: Decomposition of 45-minute transit access to full-line grocery stores, 2014–2025.



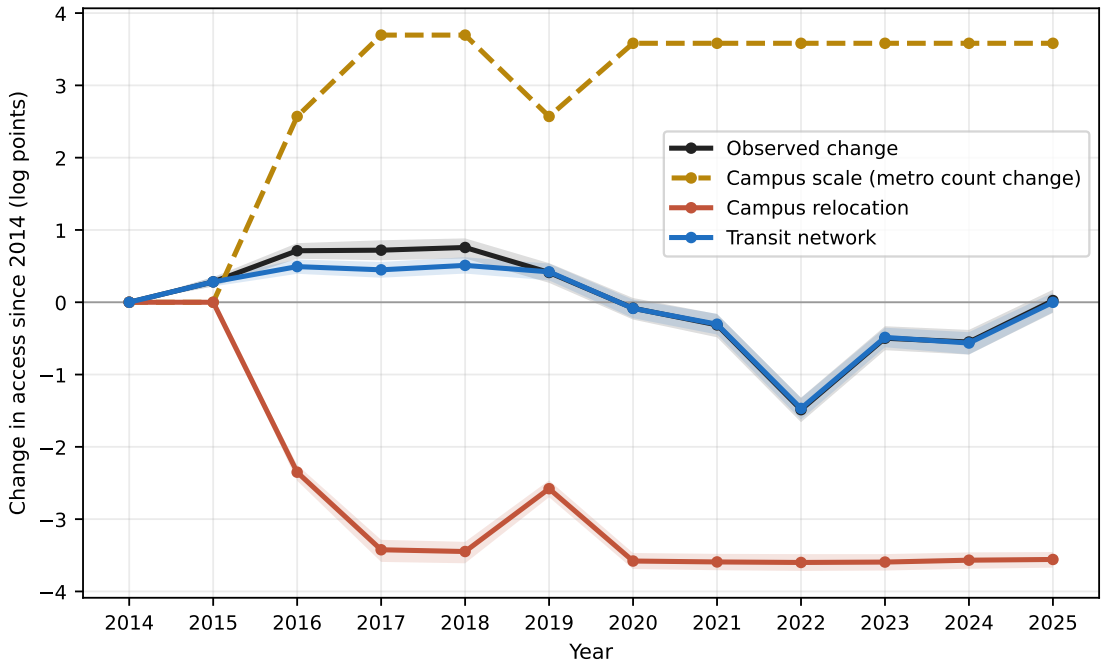
Note: Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

5.4 Community colleges

Community-college access shows near-zero net change, with modest offsetting destination components. In Figure 4 the observed 0.0 splits into 0.0 network, +3.6 campus scale, and −3.6 relocation (all-metro companion: $-0.3 = -0.3 + 1.8 - 1.8$). The relocation term mostly reflects the 87% of residents who reach no campus. Counting records rather than campuses (the persistent-identity variant), the scale and relocation terms in the all-metro companion are −8.1 and +7.9, an artifact of multi-campus systems consolidating into single records (+2.8 and −2.7 in the clean-archive core).

Campus access on the panel was nearly unchanged, at −0.4% [−0.6, −0.3] (flags included: +0.3%; NYC included: −0.5%; spanning zero under metro clustering, [−1.6, +0.7]). The persistent-identity variant gives −0.6%, and the annual-sector-code count gives −2.2%; neither additional decline reflects campus closures. Level exhibits for this service rely on the coverage and nearest-facility variants (Appendix Table A.5).

Figure 4: Decomposition of 45-minute transit access to community-college campuses, 2014–2025.



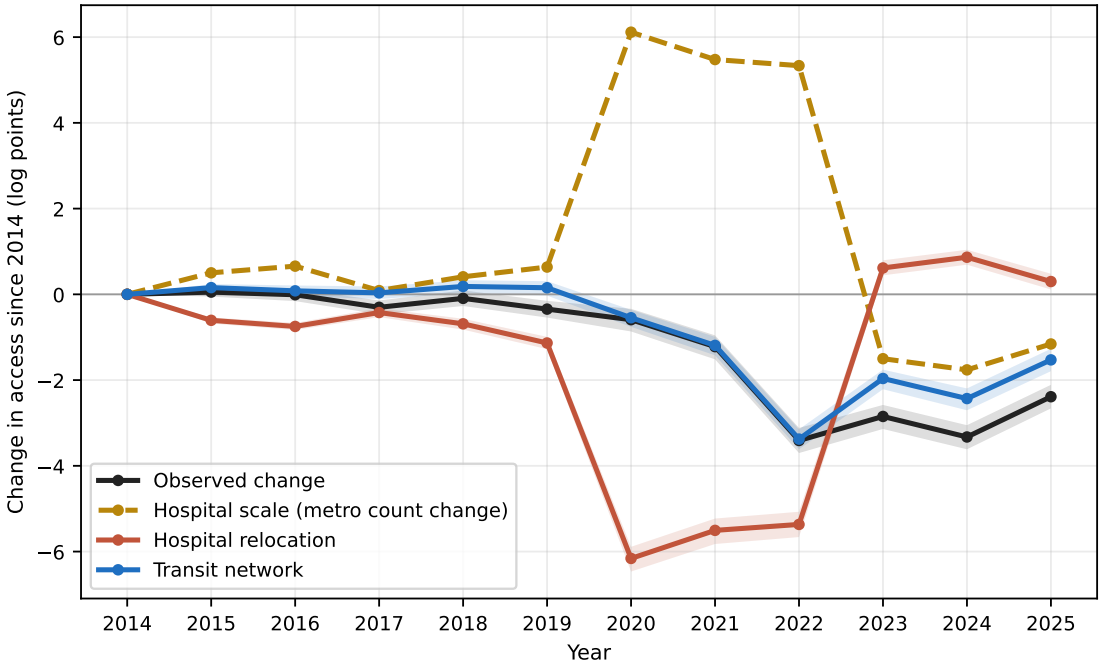
Note: Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change. Persistent-campus definition.

5.5 Hospitals

Hospital access changed through both sides, in proportions that depend on the sample. In the clean-archive core the network accounts for −1.5 of the observed −2.4, against a net destination side of −0.9 (−1.2 hospital scale, +0.3 relocation; Figure 5). In the all-metro companion the destination side is somewhat larger than the network (−1.7 against −1.4; $-3.1 = -1.4 - 4.8 + 3.0$ within rounding). The positive relocation term comes from the residents who reach no durable hospital. The opposite-signed scale and relocation movements over 2020–22 in Figure 5 are the pandemic certification cohort entering and leaving the durable set. The observed line moves little because the two components offset.

Measured as a trajectory, transit access to durable general acute care hospitals fell 3.4% [−3.6, −3.1] (flags included: −2.2%; NYC included: −3.8%), and the metro-clustered interval also excludes zero ([−5.2, −1.5]). The all-certification variant, which keeps the transient certifications, is indistinguishable at the endpoints and differs only in the 2020–22 path. The trajectory describes the roughly one quarter of residents who reach any durable hospital within 45 minutes, and the coverage and nearest-facility variants describe the rest (Appendix Table A.5).

Figure 5: Decomposition of 45-minute transit access to durable general acute hospitals, 2014–2025.



Note: Clean-archive core (28,147 block groups), population-weighted, archive-entry corrected; annual points vs. the 2014 base; shaded bands are 95% block-group-bootstrap intervals; components sum to the observed change.

6 Synthesis: mechanisms, incidence, and compounding disadvantage

6.1 One decomposition, six services

Table 3 assembles the endpoint decompositions on one scale. In the clean-archive core the network column runs from 0.0 to −6.4, while the destination columns range from −23.6 to +15.9. The all-metro companion shows the same pattern for the five services (network −0.3 to −3.2; destination columns −19.9 to +8.4; Appendix Table A.8). Destination change is a property of the service, and network change a property of the period: close to zero for every service before 2020 and negative for every service after 2019.

The same pattern holds when the clean-archive core's decompositions are split at the pandemic. Over 2014–2019 the network component lies between −0.1 and +1.1 log points for every service, while the destination columns already differ by service (bank scale −8.3, grocery scale +2.0, jobs scale +10.5). Over 2019–2025 the network component is negative for every service (−0.4 to −6.3), while the destination columns again differ by service (bank

Table 3: Cross-service endpoint decomposition.

Service	Observed	Network	Destination scale	Relocation
Jobs	+15.2	−6.4	+15.9	+5.7
Bank branches	−21.4	−4.0	−23.6	+6.2
Public libraries	−1.0	−2.5	+3.3	−1.8
Groceries (full-line)	+0.4	−2.8	+6.5	−3.3
Community colleges	0.0	0.0	+3.6	−3.6
Durable general acute hospitals	−2.4	−1.5	−1.2	+0.3

Note: 2025 vs. 2014 annual average; clean-archive core; all entries in log points, so the banks observed −21.4 corresponds to −19.3 percent. Components sum to observed; block-group bootstrap 95% CIs for observed, network, and relocation are within ±1.0 of the point estimates shown. Appendix A reports the all-metro companion for every service row. The jobs row is the decomposition of Arnold (2026) on the same sample rule.

scale −15.3, grocery scale +4.5).

6.2 Who has access, and who lost it

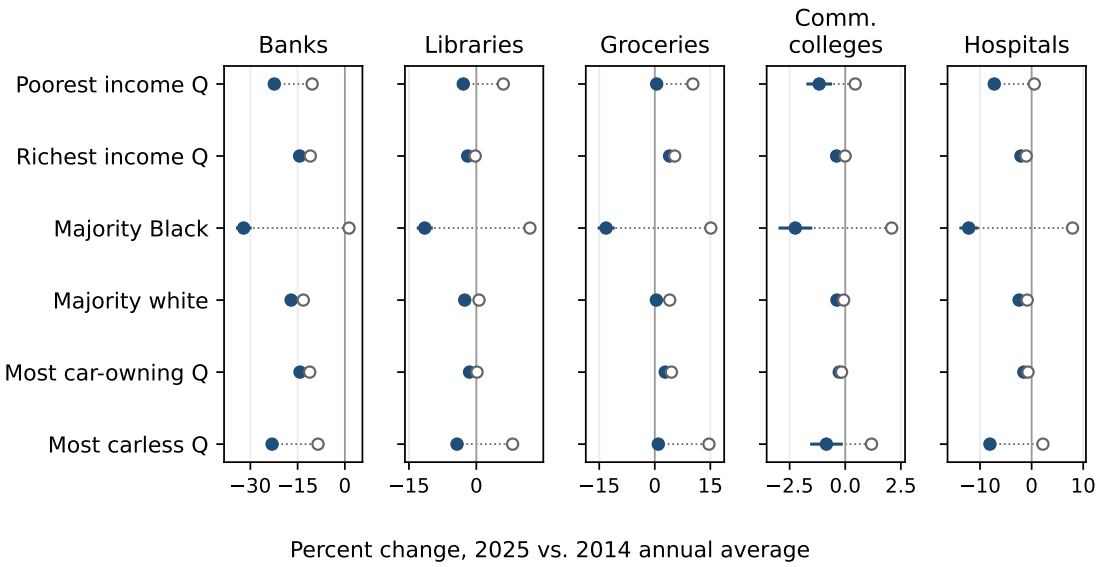
Demographic attributes are the fixed 2018–22 five-year ACS vintage; level exhibits are quoted for 2023, the last year in which every layer, jobs included, is observed rather than held. Majority-race classes are population-share majorities; income quartiles are quartiles of block-group median household income within each metro; carless quartiles are quartiles of the block-group carless-household share across the panel.

Levels. Outside New York, transit access to every essential service is higher in poor, majority-Black, and carless neighborhoods than in rich, white, car-owning ones. In 2023, richest-to-poorest income-quartile ratios run 0.35–0.53 across the six services, white-to-Black majority-neighborhood ratios run 0.35–0.56, and the most carless quartile of neighborhoods has five to nine times the access of the most car-owning quartile.

Change. Over 2014–2025 this advantage narrowed, and the incidence of the declines was regressive. On the corrected series (the filled markers in Figure 6), which we use as the incidence estimate, the largest access declines fall on majority-Black neighborhoods in every service (banks −32% against −17% in majority-white neighborhoods; libraries −11.5% against −2.6%; groceries −13.2% against +0.4%; hospitals −12.2% against −2.4%). They also fall on the poorest and most carless quartiles relative to the richest and most car-owning in nearly all services. The characterizations replicate almost exactly under beginning-of-period (2009–13) classifications (U.S. Census Bureau, 2014), mapped from 2010 to 2020 block groups with the crosswalk of the IPUMS National Historical Geographic Information

System (NHGIS; Schroeder et al., 2026; Appendix Table A.3), so the incidence does not depend on who moved in after access changed. Majority-Black neighborhoods also keep the largest corrected decline in every service when each of the five metros holding 63% of their residents is dropped in turn (Appendix Table A.3).

Figure 6: Access change by neighborhood group and service, corrected and uncorrected.



Note: Percent change in the 45-minute count, 2025 vs. the 2014 base; the panel (52 metros ex-NYC), population-weighted, flagged metro-quarters excluded; filled = corrected with 95% block-group-clustered CIs, open = uncorrected upper bound; panel x-scales differ by service.

Measurement. For advantaged groups the corrected and uncorrected estimates differ by at most 3.9 percentage points (majority-white banks). For disadvantaged groups they diverge by up to 33 percentage points (majority-Black banks: −32.1 to +1.3). Archive coverage changes concentrate in the agencies serving disadvantaged areas, so the differences between groups are sensitive to the archive-entry correction in a way the aggregate results are not. The width of the majority-Black bank bracket is traceable to Philadelphia, whose Southeastern Pennsylvania Transportation Authority (SEPTA) feeds entered the archive mid-panel and which supplies 84% of the archive-entry correction carried by majority-Black block groups. Excluding Philadelphia moves the corrected majority-Black bank change only from −32.1 to −30.3 percent, and the corrected majority-white change from −17.1 to −16.9, but moves the uncorrected majority-Black change from +1.3 to −18.6 percent. The corrected series is the one stable to it (Appendix Table A.3), so we quote group-level changes as bounds and report the asymmetry as a result.

6.3 Compounding disadvantage

We call a neighborhood a loser of service s if its within-neighborhood corrected access change from the 2014 base to 2025 is -5 log points or worse. We then compare multi-service loss rates to the benchmark in which each service's losses (at their observed marginal rates) occur independently. Shares and multipliers count block groups unweighted. Loss marginals differ widely by service (banks 55%, jobs 31%, groceries 28%, libraries 17%, hospitals 11%, community colleges 3%).

Table 4 reports the result. Loss in two or more services occurs below the independence rate (×0.84), and the multiplier rises from ×1.6 at three services to ×31.5 at five or more. Four checks aimed at possible artifacts leave the rise in place: (i) against each metro's own loss marginals, which absorbs between-metro composition, ×3.5 at four or more services and ×13 at five or more; (ii) under population weighting, ×6.9 and ×44; (iii) dropping community colleges (the sparsest layer) or jobs entirely, ×6.2 and ×41 without community colleges and ×13 and ×57 without jobs (at four or more and five or more of the remaining five services); (iv) dropping Seattle, Chicago, and Denver, the three metros that hold most of the neighborhoods that lost all six services, ×9.6 and ×73. Without the three core metros whose archived feeds omit rail in some quarters (Section 3), the multipliers are ×5.0 and ×25.4 (Appendix Table A.3). In the all-metro companion the multipliers for four or more and for five or more services are ×5.3 and ×28.6.

Table 4: Multi-service loss against the independence benchmark.

Services lost	Observed	Indep.	Mult.	Metro 95% CI	Within strata	Within metro
≥ 2 of 6	38.6%	45.7%	×0.84	—	—	—
≥ 3 of 6	22.3%	13.9%	×1.6	—	—	—
≥ 4 of 6	11.9%	2.15%	×5.5	[3.7, 9.8]	×1.08	×3.5
≥ 5 of 6	4.63%	0.147%	×31.5	[16.2, 76.4]	×1.33	×13.0
All six	0.455%	0.0029%	×157	[65.0, 580.9]	×1.47	×40.4

Note: Clean-archive core. Losses: corrected 2014–2025 change ≤ −5 log points. Strata = the block groups whose mean network component across the six services is exactly zero (36%), and quartiles of the rest; within-metro = each metro's own loss marginals. Conditional multipliers pool observed over expected losers across strata or metros, each expected count from that group's own loss marginals. Metro-clustered intervals are percentile bootstraps over metros (500 draws); they and the conditional multipliers are computed for the deep thresholds (≥ 4) only. Indep. = the share expected if losses were independent; Mult. = observed over independent.

Holding the network component fixed removes most of the rise. Within network strata, which group block groups by their mean network component across the six services and so absorb the component the services share, the multipliers fall to ×1.08 at four or more services (95% metro-clustered interval 0.99 to 1.14) and ×1.33 at five or more (1.17 to 1.42). Jointly within metro-by-stratum cells they fall to ×1.01 and ×1.19.

The 128 neighborhoods (0.45%) that lost all six services are ×157 over independence unconditionally and ×1.47 within network strata, and 72% of them are in three metros (Seattle, Chicago, Denver); Seattle's share partly reflects the Link gap in its archived feeds (Section 3). Losing several services at once therefore mostly indicates network decline rather than destinations closing independently in the same places. The multipliers are larger under a stricter −10 log-point definition (×6.5 at four or more services and ×40 at five or more).

The network reading is consistent with agency-reported service. Across the 19 core metros, the banks network component correlates at $r = 0.50$ with the change in National Transit Database (NTD) vehicle revenue miles (Federal Transit Administration, 2026; Appendix Table A.3). The check is not fully independent of the sample, because disagreement with these revenue miles is one of the criteria that flag archive quarters (Appendix Table A.6).

From neighborhoods losing no service to those losing all six, the low-income population share rises from .21 to .29, the carless household share from .04 to .12, the Black population share from .06 to .13, and majority-minority status from 34% to 44%. Median household income falls from $106,000 to $89,000, and distance to the metro center falls from 31 to 18 km. The carless share and distance to the metro center change monotonically through four services lost, and the low-income share, Black share, and median income through three. Beyond those points the gradients level off or partly reverse; counts are small at five and six services. The pattern replicates under beginning-of-period (2009–13) classifications (low-income share .25 to .39, carless .04 to .13, median income $75,000 to $56,000).

7 New York: the exception

New York is excluded from the paper's samples and reported here instead. Four facts justify the treatment.

Scale. New York contributes 14,472 every-year block groups (19.8 million residents), half the size of the clean-archive core (28,147 block groups; 41.8 million residents). Pooled population-weighted results would give New York about a third of the weight.

Grocery and hospital access change more. In a single metro every block group changes its set of archived feeds in the same quarters, so the trajectory estimator cannot separate an archive-entry step from a real change in those quarters. New York's decompositions are reported uncorrected to match its trajectories, and the only like-for-like comparison is between New York's uncorrected changes and the clean-archive core's uncorrected changes

(Table 5). New York's grocery access rose 10.1 log points against 0.7 in the core, and its hospital access fell 11.7 against 2.3, while bank access changed by similar amounts in both.

Table 5: New York against the clean-archive core, like-for-like.

Service	New York	Clean-archive core
Bank branches	−19.7	−20.9
Public libraries	−2.6	−0.5
Groceries (full-line)	+10.1	+0.7
Community colleges	−0.8	0.0
Durable general acute hospitals	−11.7	−2.3

Note: Uncorrected endpoint change in the 45-minute count, 2025 vs. the 2014 annual average, log points; New York = 14,472 every-year block groups; population-weighted.

The decompositions are more destination-driven. The bank decomposition shows the core's pattern in stronger form. The network contributes essentially nothing (+0.7), the branch decline is steeper (−28.3 against −23.6), and the relocation term is larger (+7.9 against +6.2). The grocery gain is nearly all store growth (+12.1 scale against −2.0 relocation), and the library decline comes from location rather than the network (−2.0 relocation, network ≈ 0).

The race gradient reverses for some services. In New York the income and race gradients are flatter than in the panel, and the race gradient reverses sign for jobs, banks, and community colleges (white/Black 1.08–1.26; the jobs and banks intervals include parity). Any pooled "who has access" exhibit would average opposite-signed race gradients for those services.

8 Conclusion

Between 2014 and 2025, transit access to essential services in the 52 metros outside New York changed in different directions and for different reasons. Branch closures drove the bank decline and the network drove the library decline; hospitals lost on both sides, grocery store growth offset network decline, and community colleges were nearly unchanged (Table 3). Only the bank and hospital declines are distinguishable from zero when the intervals are clustered by metro. Losses in several services at once were far more common than independent declines would imply, mostly because the transit network the services share declined in the same neighborhoods (Table 4).

Every service's network component was close to zero before 2020 and negative after 2019, while the destination side followed each service's own industry: branch consolidation, supermarket growth, and hospital openings and closures. The declines also differed across neighborhoods. Under the archive-entry correction, the largest declines in every essential service were in majority-Black neighborhoods. The poorest and most carless neighborhoods had larger declines than the richest and most car-owning in nearly all services. These are also the neighborhoods whose access history is hardest to measure (Section 6.2).

The results have three implications for practice. First, the decomposition identifies which institution can act. Where the network component dominates, as for libraries and where the network declined, the relevant policy is transit service. Where the destination side dominates, as for banks, transit service cannot reverse branch closures, though it can restore access to the branches that remain. Siting and community-reinvestment review could target the closures that leave a neighborhood with no reachable branch (Appendix Table A.5). Second, access to essential services can be monitored with public data. The inputs are agencies' published schedules and federal facility files, and the travel-time matrices that measure job access serve every service without additional routing. Third, the neighborhoods losing many essentials at once are identifiable, and they are poorer and more carless than the neighborhoods that lost none (Section 6.3).

The measure has limits. It is potential access and does not account for facility capacity, service quality, or frequency and transfer burden beyond their effect on median travel time. It is computed from published schedules, so it measures scheduled service, which can differ from delivered service and from the reliability a rider experiences. It is measured at the morning peak (Section 4) on an archive that favors larger agencies (Section 3.3), and destination vintages are annual, so within-year destination movement is not observed. At two of the twelve archive-entry events (Appendix Table A.3), agency-reported service moved by 5 log points or more in the same direction as archive coverage: an expansion in Santa Rosa in 2017, where the archive-entry correction may remove some real gain, and a contraction in Sacramento in 2020, whose comparison window includes the April 2020 pandemic service cut and the start of the light-rail gap in its archived feeds.

The analysis is descriptive. It records how access changed and which side moved, not whether the changes affected banking, food security, health, or education. For studies of those outcomes, the panel provides a neighborhood-level measure of access to five services that is comparable over time.

Neither transit service nor facility counts alone would have shown both the branch closures and the network losses that removed several services at once from the neighborhoods that depend most on transit.

References

Robert M. Adams and Shane M. Sherlund. Where's the bank? Banking access in the era of branch consolidation. Technical Report 2025-086, Federal Reserve Board, 2025.

Jeff Allen and Steven Farber. Planning transport for social inclusion: An accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment*, 78:102212, 2020. doi: 10.1016/j.trd.2019.102212.

Sierra Arnold. Public transportation access and food insecurity. *Journal of Regional Science*, 65(5):1449–1467, 2025. doi: 10.1111/jors.70010.

Sierra Arnold. The evolution of transit access to jobs in American cities. Working paper, Xavier University, 2026.

Geneviève Boisjoly and Ahmed El-Geneidy. Daily fluctuations in transit and job availability: A comparative assessment of time-sensitive accessibility measures. *Journal of Transport Geography*, 52:73–81, 2016. doi: 10.1016/j.jtrangeo.2016.03.004.

R. Blake Buchalter, Paul R. Gunsalus, Madeleine M. Blazel, Michael W. Kenyhercz, and Jarrod E. Dalton. Disparities in multi-modal spatial access to primary and specialty care in U.S. neighborhoods: A cross-sectional and temporal analysis of health services. *PLoS One*, 20(9):e0330427, 2025. doi: 10.1371/journal.pone.0330427.

Centers for Medicare and Medicaid Services. Provider of Services File (Hospital and Non-Hospital Facilities), December Vintages 2013–2025 [dataset]. data.cms.gov, Office of Enterprise Data and Analytics, 2026. URL https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/provider-of-services-file-quality-improvement-and-evaluation-system. Now titled "Provider of Services File – Quality Improvement and Evaluation System." Accessed July 10, 2026.

M. W. Conway, A. Byrd, and M. van Eggermond. Accounting for uncertainty and variation in accessibility metrics for public transport sketch planning. *Journal of Transport and Land Use*, 11(1):541–558, 2018. doi: 10.5198/jtlu.2018.1074.

Mengying Cui and David Levinson. Primal and dual access. *Geographical Analysis*, 52(3): 452–474, 2020. doi: 10.1111/gean.12220.

Department for Transport. Journey Time Statistics, England: 2019. Official statistics, Department for Transport, 2021. URL

https://www.gov.uk/government/statistics/journey-time-statistics-england-2019/journey-time-statistics-england-2019. Published 4 November 2021. Accessed November 11, 2026.

Francis P. Donnelly. Regional variations in average distance to public libraries in the United States. *Library & Information Science Research*, 37(4):280–289, 2015. doi: 10.1016/j.lisr.2015.11.008.

Alireza Ermagun and Frank Witlox. Transit access effectiveness in American metropolitan areas. *Journal of Transport Geography*, 116:103853, 2024. doi: 10.1016/j.jtrangeo.2024.103853.

Alireza Ermagun, Fatemeh Janatabadi, and Frank Witlox. Beyond the 15-minute city dichotomy: Time-denominated access to essential services in Chicago. *Transportation Research Part A: Policy and Practice*, 195:104428, 2025. doi: 10.1016/j.tra.2025.104428.

Steven Farber and Liwei Fu. Dynamic public transit accessibility using travel time cubes: Comparing the effects of infrastructure (dis)investments over time. *Computers, Environment and Urban Systems*, 62:30–40, 2017. doi: 10.1016/j.compenvurbsys.2016.10.005.

Steven Farber, Melinda Z. Morang, and Michael J. Widener. Temporal variability in transit-based accessibility to supermarkets. *Applied Geography*, 53:149–159, 2014. doi: 10.1016/j.apgeog.2014.06.012.

Federal Deposit Insurance Corporation. Summary of Deposits, 2013–2025 [dataset]. FDIC BankFind Suite, 2025. URL https://banks.data.fdic.gov/bankfind-suite/SOD. Annual survey of branch office deposits as of June 30. Accessed July 8, 2026.

Federal Transit Administration. National Transit Database, Complete Monthly Ridership (with Adjustments and Estimates) [dataset]. U.S. Department of Transportation open data portal, dataset 8bui-9xvu, 2026. URL https://data.transportation.gov/d/8bui-9xvu. Monthly vehicle revenue miles by agency, mode, and type of service. Accessed July 6, 2026.

Nicole Foth, Kevin Manaugh, and Ahmed El-Geneidy. Towards equitable transit: Examining transit accessibility and social need in Toronto, Canada, 1996–2006. *Journal of Transport Geography*, 29:1–10, 2013. doi: 10.1016/j.jtrangeo.2012.12.008.

Karst T. Geurs and Bert van Wee. Accessibility evaluation of land-use and transport strategies: Review and research directions. *Journal of Transport Geography*, 12(2): 127–140, 2004. doi: 10.1016/j.jtrangeo.2003.10.005.

Joe Grengs. Nonwork accessibility as a social equity indicator. *International Journal of Sustainable Transportation*, 9(1):1–14, 2015. doi: 10.1080/15568318.2012.719582.

Susan L. Handy and Debbie A. Niemeier. Measuring accessibility: An exploration of issues and alternatives. *Environment and Planning A*, 29(7):1175–1194, 1997. doi: 10.1068/a291175.

Walter G. Hansen. How accessibility shapes land use. *Journal of the American Institute of Planners*, 25(2):73–76, 1959. doi: 10.1080/01944365908978307.

Nicholas W. Hillman. Geography of college opportunity: The case of education deserts. *American Educational Research Journal*, 53(4):987–1021, 2016. doi: 10.3102/0002831216653204.

Homeland Infrastructure Foundation-Level Data. Hospitals [dataset]. HIFLD Open hospitals layer, copy hosted on ArcGIS Online, 2020. URL https://www.arcgis.com/home/item.html?id=fbd736551b834a069258dda168b7c95f. 7,581 facilities; source dates through August 13, 2019. Accessed July 13, 2026.

Institute of Museum and Library Services. Public Libraries Survey, Public Library Outlet Data Files, Fiscal Years 2013–2024 [dataset]. Institute of Museum and Library Services, Washington, DC, 2026. URL https://www.imls.gov/research-evaluation/surveys/public-libraries-survey-pls. Accessed July 9, 2026.

Fatemeh Janatabadi and Alireza Ermagun. Unraveling transit service and land use components of the socio-spatial inequality of access. *Environment and Planning B: Urban Analytics and City Science*, 51(7):1447–1462, 2024a. doi: 10.1177/23998083231207534.

Fatemeh Janatabadi and Alireza Ermagun. Four-step framework for untangling place and mobility components of access: An equal opportunity perspective. *Journal of Transport Geography*, 116:103831, 2024b. doi: 10.1016/j.jtrangeo.2024.103831.

Javad Jomehpour Chahar Aman and Janille Smith-Colin. Transit deserts: Equity analysis of public transit accessibility. *Journal of Transport Geography*, 89:102869, 2020. doi: 10.1016/j.jtrangeo.2020.102869.

Armita Kar, Andre L. Carrel, Harvey J. Miller, and Huyen T. K. Le. Public transit cuts during COVID-19 compound social vulnerability in 22 US cities. *Transportation Research Part D: Transport and Environment*, 110:103435, 2022. doi: 10.1016/j.trd.2022.103435.

Alex Karner. Assessing public transit service equity using route-level accessibility measures and public data. *Journal of Transport Geography*, 67:24–32, 2018. doi: 10.1016/j.jtrangeo.2018.01.005.

Keone Kelobonye, Gary McCarney, Jianhong (Cecilia) Xia, Mohammad Shahidul Hasan Swapan, Feng Mao, and Heng Zhou. Relative accessibility analysis for key land uses: A spatial equity perspective. *Journal of Transport Geography*, 75:82–93, 2019. doi: 10.1016/j.jtrangeo.2019.01.015.

Willem Klumpenhouwer, Jeff Allen, Lisa Li, Rick Liu, Mitchell Robinson, Diego Da Silva, Steven Farber, Alex Karner, Dana Rowangould, Amer Shalaby, Mary Buchanan, and Steven Higashide. A comprehensive transit accessibility and equity dashboard. *Findings*, July 2021. doi: 10.32866/001c.25224.

Alexander Li, Aurimas Racas, Junaid Syed, Mengyang Liu, Przemek Zientala, and Tejas Santanam. Ease and equity of point of interest accessibility via public transit in the U.S. In *2025 IEEE International Conference on Future Machine Learning and Data Science (FMLDS)*, pages 519–523. IEEE, 2025. doi: 10.1109/FMLDS67896.2025.00132.

Karen Lucas. Transport and social exclusion: Where are we now? *Transport Policy*, 20: 105–113, 2012. doi: 10.1016/j.tranpol.2012.01.013.

Wei Luo and Fahui Wang. Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design*, 30(6):865–884, 2003. doi: 10.1068/b29120.

Karel Martens. Justice in transport as justice in accessibility: Applying Walzer's 'Spheres of Justice' to the transport sector. *Transportation*, 39(6):1035–1053, 2012. doi: 10.1007/s11116-012-9388-7.

National Center for Education Statistics. Integrated Postsecondary Education Data System, Institutional Characteristics: Directory Information Files, 2013–2024 [dataset]. IPEDS Data Center, U.S. Department of Education, 2025. URL `https://nces.ed.gov/ipeds/datacenter/DataFiles.aspx`. Files HD2013–HD2024. Accessed July 9, 2026.

Tijs Neutens. Accessibility, equity and health care: Review and research directions for transport geographers. *Journal of Transport Geography*, 43:14–27, 2015. doi: 10.1016/j.jtrangeo.2014.12.006.

Andrew Owen, Shirley Shiqin Liu, and Eric M. Lind. Access across America: Transit 2024. Technical Report CTS 25-18, University of Minnesota Accessibility Observatory, 2025. URL https://hdl.handle.net/11299/277743. Accessed November 11, 2026.

Rafael H. M. Pereira, Tim Schwanen, and David Banister. Distributive justice and equity in transportation. *Transport Reviews*, 37(2):170–191, 2017. doi: 10.1080/01441647.2016.1257660.

Pew Research Center. 1 in 10 Americans rarely or never drive a car, 2024. URL https://www.pewresearch.org/short-reads/2024/11/14/1-in-10-americans-rarely-or-never-drive-a-car/. Short Read, November 14 (nationally representative survey of 5,410 U.S. adults, August 12–18, 2024). Accessed November 11, 2026.

Piotr Rosik, Marcin Mazur, Marcin Stępniak, Tomasz Komornicki, Sławomir Goliszek, Patryk Duma, and Paweł Churski. What drives potential accessibility decomposition? *Journal of Transport and Land Use*, 18(1):1–27, 2025. doi: 10.5198/jtlu.2025.2508.

Manuel Santana Palacios and Ahmed El-Geneidy. Cumulative versus gravity-based accessibility measures: Which one to use? *Findings*, February 2022. doi: 10.32866/001c.32444.

Jonathan Schroeder, David Van Riper, Steven Manson, Grace Cooper, Zachary Krause, Tracy Kugler, Tsu Zhu, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version 21.0 [dataset]. Minneapolis, MN: IPUMS, 2026.

Anthony F. Shorrocks. Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *Journal of Economic Inequality*, 11(1):99–126, 2013. doi: 10.1007/s10888-011-9214-z.

Sei-Ching Joanna Sin. Neighborhood disparities in access to information resources: Measuring and mapping U.S. public libraries' funding and service landscapes. *Library & Information Science Research*, 33(1):41–53, 2011. doi: 10.1016/j.lisr.2010.06.002.

Marcin Stępniak and Piotr Rosik. The role of transport and population components in change in accessibility: The influence of the distance decay parameter. *Networks and Spatial Economics*, 18(2):291–312, 2018. doi: 10.1007/s11067-017-9376-8.

U.S. Census Bureau. American Community Survey, 5-Year Estimates, 2009–2013: Detailed Tables [dataset]. Census Data API, 2014. URL https://api.census.gov/data/2013/acs/acs5. Released December 4, 2014. Accessed November 11, 2026.

U.S. Census Bureau. American Community Survey, 5-Year Estimates, 2018–2022: Detailed Tables [dataset]. Census Data API, 2023a. URL https://api.census.gov/data/2022/acs/acs5. Released December 7, 2023. Accessed November 11, 2026.

U.S. Census Bureau. American Community Survey, 1-Year Estimates [dataset], 2023b. Table B25044, Tenure by Vehicles Available.

U.S. Census Bureau. County Business Patterns: 2023 [dataset]. U.S. Census Bureau, CBP Datasets, 2025. URL https://www.census.gov/data/datasets/2023/econ/cbp/2023-cbp.html. Released June 26, 2025. Accessed November 11, 2026.

U.S. Census Bureau. LEHD Origin-Destination Employment Statistics, Version 8: Workplace Area Characteristics Files, 2014–2023 [dataset], 2026. URL https://lehd.ces.census.gov/data/lodes. Accessed June 13, 2026.

U.S. Department of Agriculture, Food and Nutrition Administration. SNAP Retailer Locator Historical Data, 2005–2025 [dataset]. USDA Food and Nutrition Administration, 2026. URL https://www.fna.usda.gov/snap/retailer-locator/data. Data current as of December 31, 2025. Accessed July 9, 2026.

Michele Ver Ploeg, Vince Breneman, Tracey Farrigan, Karen Hamrick, David Hopkins, Phillip Kaufman, Biing-Hwan Lin, Mark Nord, Travis A. Smith, Ryan Williams, Kelly Kinnison, Carol Olander, Anita Singh, and Elizabeth Tuckermanty. Access to affordable and nutritious food: Measuring and understanding food deserts and their consequences. Report to Congress, Administrative Publication 036, USDA Economic Research Service, 2009.

Martin Wachs and T. Gordon Kumagai. Physical accessibility as a social indicator. *Socio-Economic Planning Sciences*, 7(5):437–456, 1973. doi: 10.1016/0038-0121(73)90041-4.

Michael J. Widener, Steven Farber, Tijs Neutens, and Mark Horner. Spatiotemporal accessibility to supermarkets using public transit: An interaction potential approach in

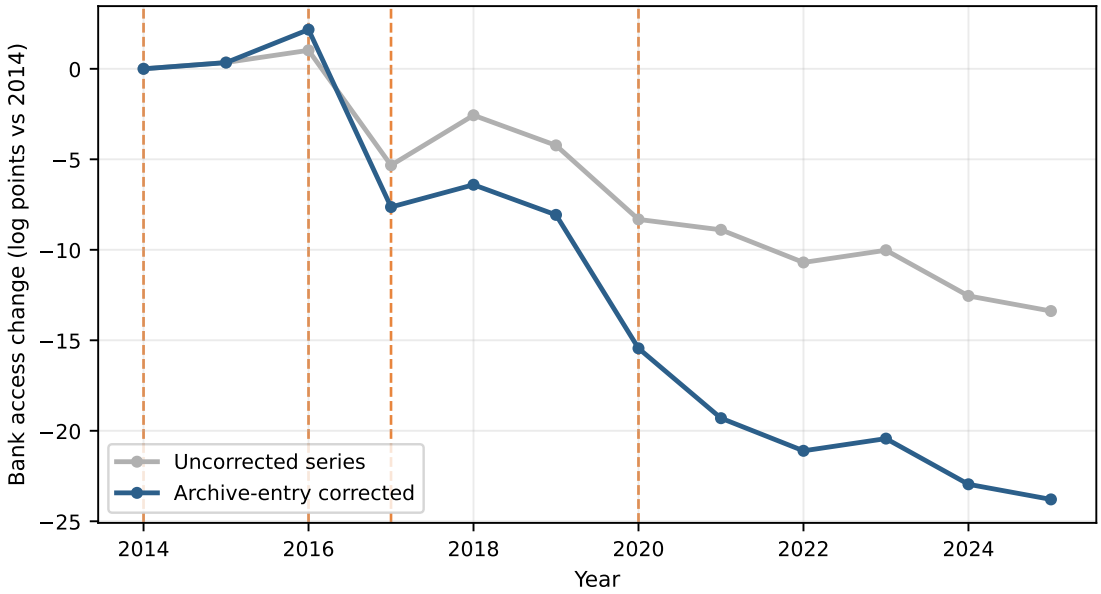
Cincinnati, Ohio. *Journal of Transport Geography*, 42:72–83, 2015. doi: 10.1016/j.jtrangeo.2014.11.004.

Y. Zhang, F. Xu, E. Talen, M. C. González, and Y. Li. Quantifying racial inequality in transit access across New York City. *PNAS Nexus*, 5(5):pgag025, 2026. doi: 10.1093/pnasnexus/pgag025.

Appendix A Supporting exhibits

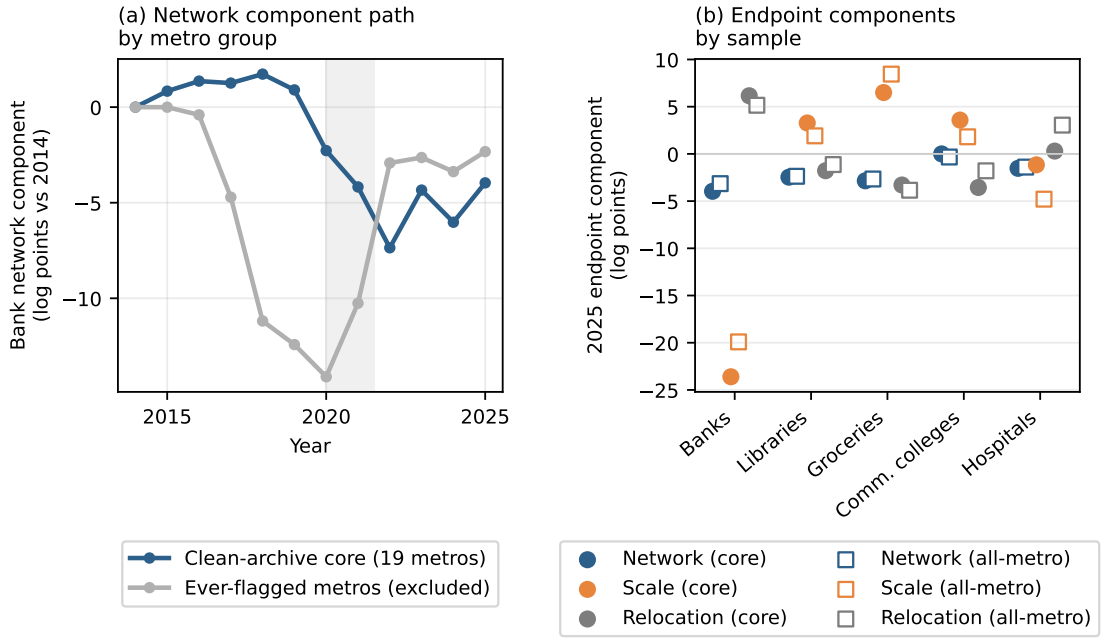
Figures

Figure A.1: The archive-entry correction in one metro (Cincinnati, bank access).



Note: Uncorrected and corrected series, log points vs. 2014; vertical marks = changes in the metro's set of archived feeds, dated from the feed inventory of Arnold (2026) (one to ten per metro); population-weighted every-year block groups. For each series, each block group's cumulative step at these changes since its 2014 base is removed from the observed and network sides equally, preserving exact additivity. Corrections are nonzero for 2–27% of block-group years, depending on the service. Validation against agency-reported revenue miles is the archive-entry row of Table A.3.

Figure A.2: Component paths by sample.



Note: Panel (a) shows the banks network component, clean-archive core vs. excluded metros; panel (b) shows the 2025 endpoint components, clean-archive core vs. all-metro companion. Both panels are population-weighted, corrected, log points vs. 2014. Sample definitions. The panel: the balanced set of 52 metros outside New York with at least 36 of 48 quarters routed; all trajectory and group-level estimates. The clean-archive core: the panel's every-year block groups (those observed and unflagged in every year) in the 19 never-flagged metros observed in every year from 2014 to 2025, 28,147 block groups, constant by construction; the same rule defines the jobs decomposition of Arnold (2026). The all-metro companion: the panel's every-year block groups in every metro that has them, which is 35 of the 52 metros (the 19 core metros and 16 ever-flagged metros; 55,359 block groups). The core holds 32.5% of the panel's population; its roster is Table A.1, and Table A.8 reports both samples' decompositions. Metros with any flagged quarter are excluded from the core (Table A.6).

Tables

Table A.1: Panel metros: summary.

	Metros
Routed	96
In the analysis panel	92
With merged census attribute values	90
Balanced panel (ex-NYC; $\geq$36 of 48 quarters)	52
Clean-archive core (never flagged, observed 2014–2025)	19

Note: The 92-metro roster of the analysis panel (quarters covered, block groups, population) is Table S1 of the online supplementary material. The clean-archive core comprises Albuquerque, Austin, Champaign–Urbana, Charleston, Chicago, Cincinnati, Denver, Kansas City, Madison, Oxnard, Phoenix, Portland (OR), Rochester, Sacramento, San Antonio, Santa Rosa, Seattle, Spokane, and Stockton. Four routed metros (Fairbanks, Jacksonville, Palm Bay, Riverside–San Bernardino) are absent from the analysis panel, which requires car-travel matrices for every metro; the two Connecticut metros are in the analysis panel, but their attribute values could not be matched.

Table A.2: Hospital layer: active facilities and geocode coverage by year.

Year	Active hospitals	Geocoded	Coverage
2013	7,193	6,565	91.3%
2014	7,181	6,551	91.2%
2015	7,168	6,542	91.3%
2016	7,175	6,552	91.3%
2017	7,164	6,548	91.4%
2018	7,120	6,508	91.4%
2019	7,063	6,448	91.3%
2020	7,241	6,602	91.2%
2021	7,298	6,651	91.1%
2022	7,270	6,629	91.2%
2023	7,044	6,421	91.2%
2024	7,038	6,411	91.1%
2025	7,023	6,391	91.0%
Geocode sources (unique addresses): Census 7,684, HIFLD 789, OSM 251.			

Note: December Provider of Services vintages, category-01 facilities (hospitals, in the file's provider-category code). Three universes nest: 8,440 distinct certifications ever active as category-01 hospitals across the vintages; roughly 7,000–7,300 active in any December vintage (this table); and the durable general acute subset of Table 1 that defines the lead. Geocoding is a four-source cascade: Census batch and single-address geocoders, the Homeland Infrastructure Foundation-Level Data (HIFLD) facility layer, and OpenStreetMap (OSM).

Table A.3: Validation and sensitivity checks.

Check	Comparison	Result
Groceries vs. Census CBP 445110	state store counts, 2023	corr. 0.976 (logs); median ratio 0.78
Hospitals vs. HIFLD	state durable-GA counts, CMS December 2023 vs. HIFLD (source dates through 2019)	corr. 0.997 (logs, 56 units); median ratio 0.90
Libraries, PLS outlet vs. system files (internal consistency)	outlet counts	exact (1.000)
Bank network component vs. NTD	Δlog revenue miles, 19 core metros	corr. 0.50
Archive-entry correction vs. NTD	Δlog VRM over the 12 months either side of 12 archive-entry events (changes in the set of archived feeds that move access by $\geq$3 log points)	rank correlation -0.08 (Pearson 0.15); median $\lvert\Delta\rvert$ 2.3 vs. 3.0 log points at other metro-quarters; same direction and $\geq$5 log points at two events, Santa Rosa 2017 (expansion) and Sacramento 2020 (contraction)
Washington, DC exclusion	banks and hospital trajectories	banks unchanged ($-18.1/-18.1$); hospitals $-3.4/-3.3$
Philadelphia exclusion	majority-Black bank change, corrected vs. uncorrected	corrected -32.1 to -30.3; uncorrected $+1.3$ to -18.6
Single-metro exclusions	corrected group changes, dropping in turn Atlanta, Chicago, Washington, Miami, and Philadelphia (63% of majority-Black residents)	majority-Black decline the largest of six groups in all 25 cases; banks -28.6 to -32.2
Rail-feed gaps excluded	panel trajectories and groups without Seattle, Sacramento, Los Angeles, Dallas, and Austin; core decompositions and compounding without Seattle, Sacramento, and Austin	trajectories move $\leq$0.6 percentage points (banks -18.1); majority-Black decline still the largest in every service; mechanisms unchanged; $\times$5.0 and $\times$25.4
Group changes, 2009–13 attributes	profile and incidence contrasts	replicate (e.g. banks $-32.3/-17.3$)
Seasonality	quarter-of-year effects ex-2020/21	$\leq$0.9 log points

Note: Each row states its comparison and sample. Correlations are computed on logs; trajectory and group changes are percent, 2025 vs. the 2014 annual average, on the panel. CBP = County Business Patterns (NAICS 445110, supermarkets and other grocery stores except convenience stores); GA = general acute; CMS = Centers for Medicare and Medicaid Services; HIFLD = Homeland Infrastructure Foundation-Level Data; PLS = Public Libraries Survey; NTD = National Transit Database; VRM = vehicle revenue miles, whose changes are in log points.

Table A.4: Trajectory endpoints by travel-time budget.

Service	15 min	30 min	45 min	60 min
Banks	−5.6	−12.8	−18.1	−21.2
Libraries	−0.1	−1.0	−2.5	−4.3
Groceries	+1.6	+1.8	+1.7	+1.3
Community colleges	0.0	−0.2	−0.4	−1.0
Hospitals	−0.2	−1.2	−3.4	−5.4

Note: Percent change, 2025 vs. 2014 annual average, the panel, corrected, population-weighted.

Table A.5: Construct grid.

	Coverage: % reaching ≥1 in 45 min			Nearest facility (min)	
Service	2014	2025	Change (pp)	Median 2023	Change
Banks	75.5	72.2	−3.3	28	+2
Libraries	48.8	48.3	−0.5	48	0
Groceries	71.1	71.3	+0.2	29	0
Community colleges	10.0	10.5	+0.5	>90	—
Hospitals	24.2	22.7	−1.5	76	+2

Note: Coverage (share of population reaching at least one facility in 45 minutes, pp change) and nearest-facility (population-weighted minutes to the closest facility) variants, the panel, 2025 vs. 2014. The nearest-facility variant reads the same travel-time matrices, which hold trips of up to 90 minutes; a median beyond that is reported as >90 with no change, and nearest-facility changes are reported at nearest-minute precision.

Table A.6: Quality flags: classes, detection rules, and coverage.

Flag class	Detection rule	Cells caught	Share
Feed outage or defect	metro-quarter with a partial outage, a degraded or duplicated archived feed, or an archive loss contradicted by agency-reported revenue miles	136 of 2,330 metro-quarters	5.8%
Thin feed	routable feed with implausibly sparse service	0 beyond outages	0.0%
Metro-years touched by any flag		70 of 606	11.6%
Metros ever flagged (excluded from the core)		29	
Never flagged, missing 2014 or 2025 (excluded)		4	
Never flagged, observed 2014–2025 (the core)		19	

Note: The panel (52 metros outside New York), 2014–2025. A metro-year counts as touched when any of its quarters is flagged. Flagged metro-quarters stay in the panel for coverage and are excluded from the estimates. Rail services missing inside an unchanged set of feeds (Section 3) are not flagged.

Table A.7: Per-exhibit samples, units, and weights.

Exhibit	Sample	Units	Weights
Figs. 1–5 (decompositions)	clean-archive core (19 metros; 28,147 BGs)	log points	population
Trajectories (§5)	the panel (52 balanced metros ex-NYC)	percent	population
Fig. 6 (group bounds)	the panel	percent	population
Table 3 (cross-service)	clean-archive core	log points	population
Table A.8 (two samples)	core + all-metro companion (35 metros; 55,359 BGs; community colleges 34 metros, 55,099 BGs)	log points	population
Compounding (§6.3)	clean-archive core	BG counts	unweighted
NYC section	New York (14,472 BGs; uncorrected)	log points	population
Figs. A.1–A.2 (appendix)	core vs. excluded metros; core vs. all-metro companion; Cincinnati	log points	population

Note: BGs = block groups; the panel is defined in Section 3.3, the other samples in Section 4.

Table A.8: Endpoint decompositions under both samples.

	Clean-archive core				All-metro companion			
Service	Obs.	Net.	Scale	Reloc.	Obs.	Net.	Scale	Reloc.
Bank branches	−21.4	−4.0	−23.6	+6.2	−17.9	−3.2	−19.9	+5.2
Public libraries	−1.0	−2.5	+3.3	−1.8	−1.6	−2.4	+1.9	−1.1
Groceries (full-line)	+0.4	−2.8	+6.5	−3.3	+1.9	−2.7	+8.4	−3.8
Community colleges	0.0	0.0	+3.6	−3.6	−0.3	−0.3	+1.8	−1.8
Durable general acute hospitals	−2.4	−1.5	−1.2	+0.3	−3.1	−1.4	−4.8	+3.0

Note: 2025 vs. 2014 annual average, log points. Components sum to observed within rounding. Obs. = observed; Net. = network; Reloc. = relocation. The community-college companion row covers 34 metros (55,099 block groups), because Anchorage has no campus in some years.